GAMs: Introduction

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Introduction

Generalised additive models (GAMs) replace the strict linear predictor of GLMs with a sum of smooth functions of predictors: $g(\mathbb{E}[Y]) = \beta_0 + \sum_j s_j(X_j)$. Each s_j is a smooth, data-driven function estimated alongside the parametric components, so GAMs combine the interpretability of linear regression (additivity, coefficient-style summaries) with the flexibility of non-parametric smoothers. They are the standard tool whenever a continuous predictor's effect is suspected to be non-linear and theoretical grounds for a specific functional form are absent.

Prerequisites

A working understanding of generalised linear models, basis functions for smooth curves (B-splines, thin-plate splines), and the bias-variance trade-off that motivates smoothing penalties.

Theory

Each smooth function s_j is represented as a linear combination of basis functions with a smoothing penalty controlling wiggliness. `mgcv::gam()` defaults to penalised thin-plate regression splines, with the smoothing parameter selected automatically by REML (recommended) or generalised cross-validation (GCV). The effective degrees of freedom (edf) for each smooth quantify fitted complexity: edf $= 1$ means the smooth has reduced to a straight line, higher edf indicates more curvature. The penalty trades fit against smoothness and is what prevents over-fitting at high basis dimensions.

For categorical predictors, factor effects enter as ordinary parametric terms; for interactions between continuous predictors, tensor product smooths (`te()`) or `ti()` for pure interactions are available.

Assumptions

Standard GLM assumptions on the conditional distribution of the outcome; smooth effects are additive (no automatic interaction detection); the basis dimension is large enough to capture the underlying smoothness (`gam.check()` diagnoses this).

R Implementation

```r
library(mgcv)

set.seed(2026)
d <- data.frame(
  x = runif(200, 0, 10),
  z = rnorm(200)
)
d$y <- 2 + sin(d$x) * 2 + 0.4 * d$z + rnorm(200, 0, 1)

fit <- gam(y ~ s(x) + z, data = d)
summary(fit)
plot(fit, pages = 1, shade = TRUE)
```

```r
# Check and predict
check.gam <- gam.check(fit)
predict(fit, newdata = data.frame(x = 5, z = 0))
```

Output & Results

`summary(fit)` reports parametric coefficients (with t tests) and approximate F tests for each smooth (with edf and reference degrees of freedom). `plot(fit, shade = TRUE)` displays each smooth with a shaded confidence band around the estimated curve. `gam.check()` reports residual diagnostics and the basis-dimension k-index, which flags when more basis functions are needed.

Interpretation

A reporting sentence: "A GAM with a smooth function of x (edf 8.5, $F = 22.4$, $p < 0.001$) and a linear effect of z ($\hat{\beta} = 0.41$, $p < 0.001$) recovered the simulated sinusoidal pattern in x and the additive linear contribution of z; deviance explained 76 %, $\hat{R}^2_{\text{adj}} = 0.74$." Report edf for smooths and parametric coefficients for linear terms; never quote a single p-value for a smooth without its edf.

Practical Tips

- `s(x)` uses thin-plate regression splines by default; `s(x, bs = "cr")` for cubic regression splines (faster for one-dimensional smooths); `te(x, z)` for tensor product interactions between two continuous predictors.
- The basis dimension `k` defaults work for most smooth functions; raise it if `gam.check()` reports a low k-index, lower it for parsimony.
- Use `method = "REML"` for smoothing-parameter selection; it is more reliable than the default GCV for finite samples.
- For non-Gaussian outcomes, switch the family: `gam(y ~ s(x), family = binomial)`, `family = poisson`, etc.
- `gratia::draw(fit)` produces ggplot-style smooth-effect plots with cleaner formatting than base `plot.gam()`.
- For very large datasets, use `mgcv::bam()` with `discrete = TRUE` for substantial speed-ups; the API is otherwise identical.

R Packages Used

`mgcv` for `gam()`, `bam()`, smooth bases, and REML smoothing-parameter selection; `gratia` for ggplot-style GAM diagnostics and component plots; `mgcViz` for additional GAM visualisations including QQ plots and 3D smooth surfaces; `gam` (not to be confused with `mgcv`) for the older Hastie-Tibshirani backfitting implementation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression

- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures $\rightarrow$ mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI